Predicting to Diagnose

Group 19 | Chen Wanchuan

Cat identity, visual representations, and context in cat-photo preferences

Chen Wanchuan | Yuen Lok Yan | Lu Yuqing

Chee Ching Lam | Chan Hoi Lam

Working presentation draft
Contribution statement and course reference details require team completion.

Scope: strict G2 predictive diagnosis | Frozen internal evaluation + separate exploratory development analysis

Question and motivation

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- Which representations predict unseen matchups between familiar photographs, and what do their gains or failures reveal?

- Prediction is a diagnostic instrument: compare identity, shared visual scoring, and recorded context.

- Primary target: unseen unordered photo pairs whose endpoints are fitting photos in the same fitting component.

- We do not estimate objective attractiveness, causal visual effects, or an information limit.

protocol.md; claims.csv; claims_final.csv

Data understanding: a response is not a respondent

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- 24,896 labeled training responses; 6,224 unlabeled official-test rows; 1,558 photographs.

- Each row is one response to an ordered photo pair; repeated rows and repeated pairs are not independent respondents.

- No respondent identifier is available. Background fields record gender, pet experience/contact, future pet plans, and MBTI I/E.

- Raw files stay unchanged; IDs, labels, comparison types and image identities were audited and hashed.

README.md; tables/data_audit.csv; tables/graph_summary.csv

Preparation and target-aligned split

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Partition	Responses	Pairs	Purpose
Fitting	17,421	5,634	Estimation + preprocessing
Validation	3,742	1,208	Penalties and model selection
Internal test	3,733	1,207	One frozen evaluation

Unordered pairs stay together. Shared fitting photos are intentional for G2; no G3 transfer claim.

Full and matched strict-G2 eligibility retain all held-out rows here; connected fitting graphs define support.

artifacts/split_manifest.parquet; tables/split_coverage.csv; tables/graph_summary.csv

Features: fixed extraction, fitting-only transformations

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Block	Measurement	Model dimensions
H	20 low-level image statistics	19
C	Frozen MobileNetV3-Small pooled embedding	576
Y	YOLO masks + HSV measures + CLIP hard labels	100

Imputation, scaling, category pooling and block normalization use unique fitting photos only.

All blocks passed provenance/coverage gates and the team-recorded image review.

Automated success is not ground truth. Missing measurements and no detection remain explicit.

features/manifests/*.json; artifacts/preprocessing/*.json; tables/feature_quality.csv

Methods and justification: a matched model ladder

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Model	Displayed-A logit / prediction	Question
M0	Smoothed fitted A frequency	Constant baseline
M1	side + cat A score - cat B score	Cat identity
M2	side + photo A score - photo B score	Photo detail
M3-H/C/Y	side + w * (feature A - feature B)	Shared visual scoring
M4-H/C/Y	side + (w + group deviation) * feature difference	Recorded context

Stable logistic loss + ridge; analytic gradients; recorded convergence checks; exact pooled fallback.

protocol.md; decisions.yaml; src/estimator.py; artifacts/model_selection.json

Selection and leakage safeguards

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- Tune on common strict validation; then select the shared visual block and its matched contextual extension.

- Validation selected C (CNN). H and Y contextual procedures selected their exact pooled fallback.

- Saved freeze: models, mappings, preprocessing, masks, diagnostics, Git state and hashes before test outcomes were opened.

- Important limitation: previous analyses used these labeled data. This new internal split is not certified pristine.

artifacts/model_freeze.json; artifacts/outcome_access_log.csv; artifacts/stage8_core.json

Results: cat identity has the lowest point loss

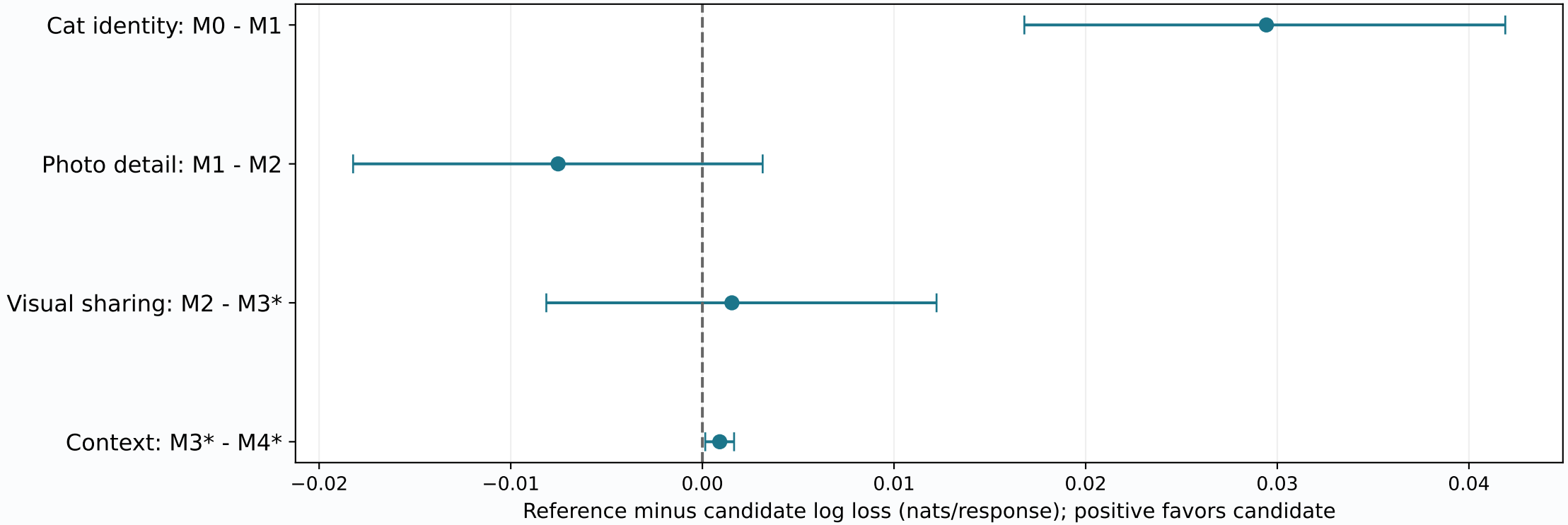
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Frozen model	Log loss	Brier	Accuracy
Constant	0.69264	0.24975	51.6%
Cat identity	0.66321	0.23537	59.2%
Photo identity	0.67074	0.23892	58.8%
Handcrafted	0.68811	0.24752	53.3%
CNN	0.66920	0.23834	58.7%
YOLO-CLIP	0.67987	0.24348	56.4%
CNN + context	0.66830	0.23789	58.9%

tables/main_results.csv | Common frozen strict-G2 test | Lower log loss is better

Evaluation: four prespecified paired contrasts

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Approximate 98.75% pointwise intervals for four primary contrasts; 5,000 shared pair resamples.

Photo detail and visual sharing span no difference; context gain is small and not causal.

tables/primary_contrasts.csv | Pair-cluster bootstrap, conditional on fitted models

Diagnosis: fitting gains need not generalize

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Model	Fitting loss	Validation loss
Cat identity	0.64618	0.66213
Photo identity	0.60792	0.66908
CNN	0.64845	0.66548

Photo-specific fitting gains did not become validation gains: consistent with estimation variance.

Preselected temperature scaling improved photo-model test loss by 0.00364 (secondary).

This supports confidence adjustment as one opportunity, not a unique explanation of remaining loss.

tables/validation_results.csv; tables/temperature_diagnostic.csv; tables/secondary_results.csv

Recorded context: small aggregate gain, not a causal effect

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- Matched CNN context gain: 0.000907 nats/response; primary interval [0.000149, 0.001654].

- Allowing group-specific side tendencies gives a similar secondary gain: 0.000895.

- The missing-context subgroup is low support and has a worse contextual point loss; aggregate improvement is not universal.

- Pet-care measurements, pair composition, reference coding and dependence remain alternative explanations.

tables/primary_contrasts.csv; tables/secondary_results.csv; tables/subgroup_results.csv

Exploratory T2: semantic choices by demographic group

Group 19 | Yuen Lok Yan

- Development-only, post-evaluation description: count choices when just one endpoint carries a semantic label.

- Example: CLIP-inferred British shorthair chosen 67.9% in current-regular-care comparisons versus 51.1% in occasional-care comparisons.

- These rates use 131 and 225 responses on different pair mixtures.

- No shared-pair group comparison passed the registered per-cell support rule. This is not an adjusted demographic preference or causal claim.

tables/exploratory_demographic_semantic_rates.csv; tables/exploratory_demographic_semantic_overlap.csv

Answer, interpretation, and unresolved limitations

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- Cat identity is useful for this known-photo unseen-pair target. The photo-detail and visual-sharing primary intervals are inconclusive.

- A small matched CNN context gain remains under the conditional bootstrap; it is predictive, not causal.

- Negative extensions are evidence: H/Y context fell back to pooling; PCA did not improve the selected CNN validation loss.

- No respondent linkage, imperfect semantic measurements, shared-image dependence, and prior analysis exposure limit conclusions.

claims_final.csv; tables/diagnostic_synthesis.csv; artifacts/model_freeze.json

Reproducibility and next evidence

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- Fresh-process refits at frozen settings reproduced 16 selected models; maximum reported test-metric difference 0.

- Fitting-only preprocessing and paired bootstrap intervals were reconstructed; no raw internal-test outcomes were reopened.

- Fixed image subsets were regenerated against H/C/Y caches. Observed differences and device/library versions are logged.

- Next evidence: independently collected pairs/respondents, stronger shared-pair context support, and validated semantic annotations.

artifacts/reproduction_statistical.json; artifacts/feature_regeneration_*.json; REPRODUCTION.md

Contributions: TEAM MUST EDIT BEFORE SUBMISSION

Group 19 | All five members

Member	Assigned responsibility (placeholder)
Chen Wanchuan	Governance reporting and reproduction
Chee Ching Lam	Data splits and joint modeling/evaluation
Lu Yuqing	H/C feature integration
Yuen Lok Yan	YOLO-CLIP integration
Chan Hoi Lam	Joint modeling and evaluation

Replace this slide with actual tasks and concrete outputs completed by each member.

AI-assisted implementation and reproduction must be reviewed by the team; assigned ownership is not proof of contribution.

tables/stage11_contributions_draft.csv | Placeholder only; not a verified record of individual work

Presentation and submission checklist

Group 19 | All five members

- Proposed substantive presenters: Chen (question/synthesis), Chee (data/evaluation), Lu (features), Yuen (YOLO-CLIP/semantics), Chan (models/uncertainty). Rehearse to approximately 15 minutes.

- Edit the contribution placeholder and complete the course Chapter 7 reference before submission.

- Submit Group_19_Report.pdf and Group_19_Code.zip locally prepared for course use only; no upload has been made.

- Official deadline: 2 November 2026, 12:00 noon HKT. Submission channel remains unverified.

Midterm_Report_Guidelines_2026.pdf; artifacts/stage11_decisions.json

References: data and methods

Group 19 | All members

- Cat Image Arena dataset and course README; missing12/cat-image-arena-images, Hugging Face.

- DSAI5101 Statistical Data Mining, Midterm Report Guidelines (2026).

- Bradley & Terry (1952), Rank Analysis of Incomplete Block Designs. Biometrika 39, 324-345.

- Wilson (1927), Probable Inference, the Law of Succession, and Statistical Inference. JASA 22, 209-212.

- Efron & Tibshirani (1993), An Introduction to the Bootstrap. Chapman & Hall.

- Course Chapter 7 materials: complete bibliographic details required from the team.

references.csv; environment lock files; feature manifests

References: representations and software

Group 19 | All members

- Howard et al. (2019), Searching for MobileNetV3. ICCV; Torchvision MobileNet_V3_Small_Weights.IMAGENET1K_V1.

- Radford et al. (2021), Learning Transferable Visual Models From Natural Language Supervision. ICML.

- OpenCLIP implementation: github.com/mlfoundations/open_clip; ViT-B-32 with OpenAI weights.

- Ultralytics YOLO11n segmentation: docs.ultralytics.com; actual checkpoint hashes are in features/manifests/Y.json.

- NumPy, SciPy, NetworkX, Matplotlib, PyArrow, Pillow, PyTorch/Torchvision, OpenCV, OpenCLIP and pypdf: exact versions in environment locks.

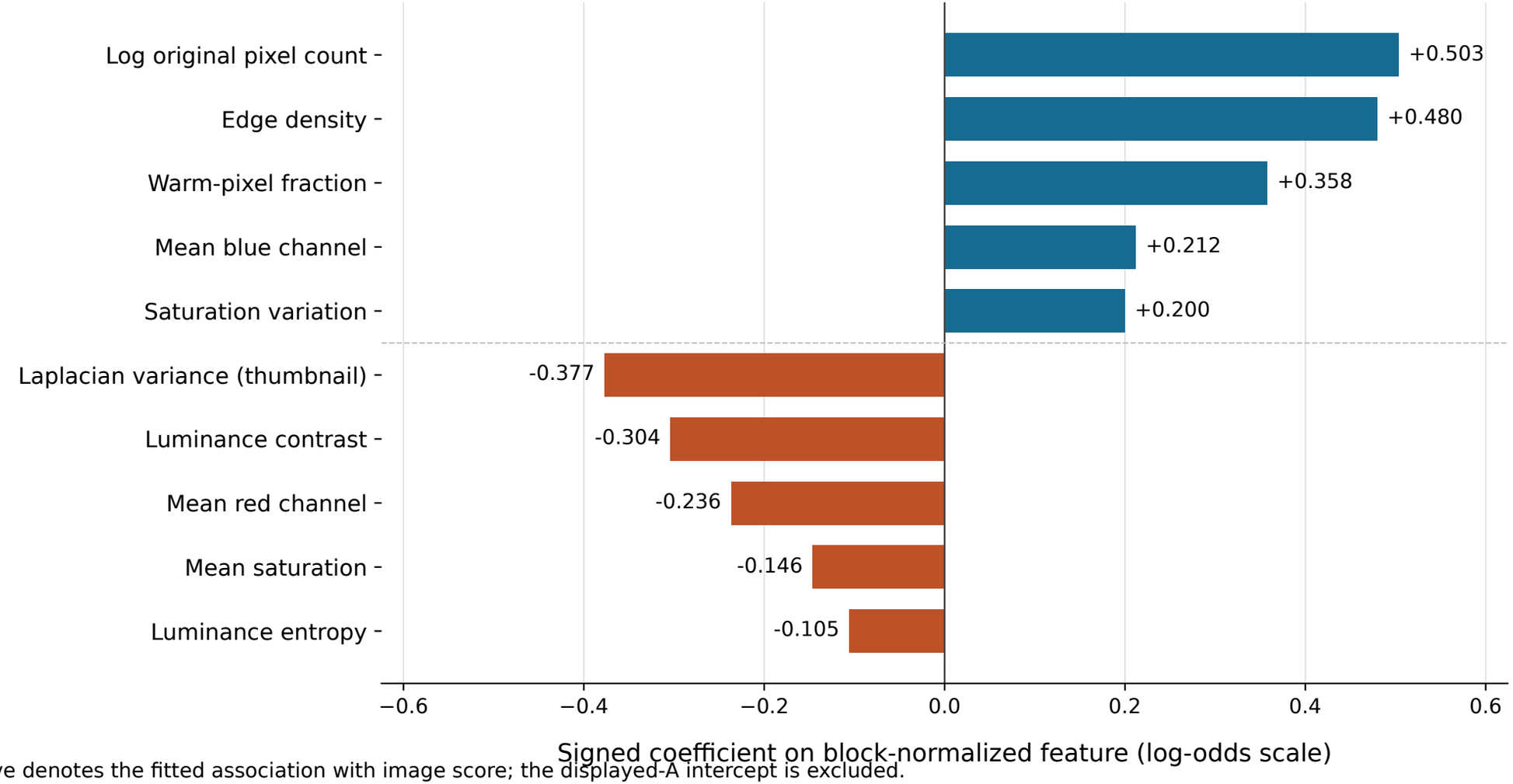
- Data/derived materials: education and research only. Weights and raw photographs are excluded from the code ZIP.

references.csv; environment lock files; feature manifests

Handcrafted image features (M3-H)

Predictive Association, Not Causal Effect

Five largest positive and five largest negative fitted feature weights



Positive / negative denotes the fitted association with image score; the displayed-A intercept is excluded.

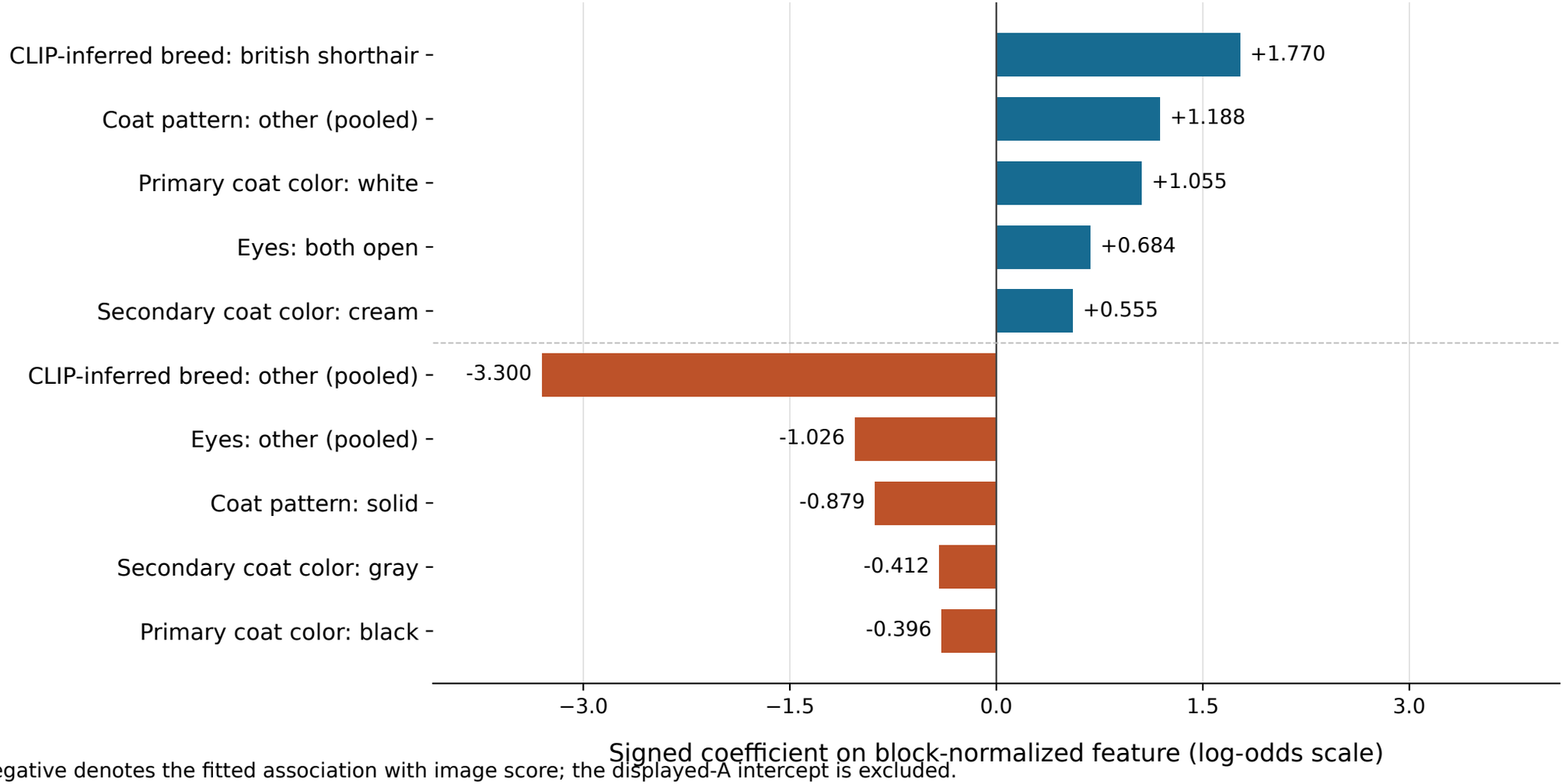
Numeric predictors use fitting-photo standardization and block normalization.

These are conditional model weights, not causal effects, effect-size uncertainty, or standalone feature importance.

Automated image attributes (M3-Y)

Predictive Association, Not Causal Effect

Five largest positive and five largest negative fitted feature weights



Positive / negative denotes the fitted association with image score; the displayed-A intercept is excluded.

Category differences must be interpreted within the same attribute; indicator weights depend on model coding.

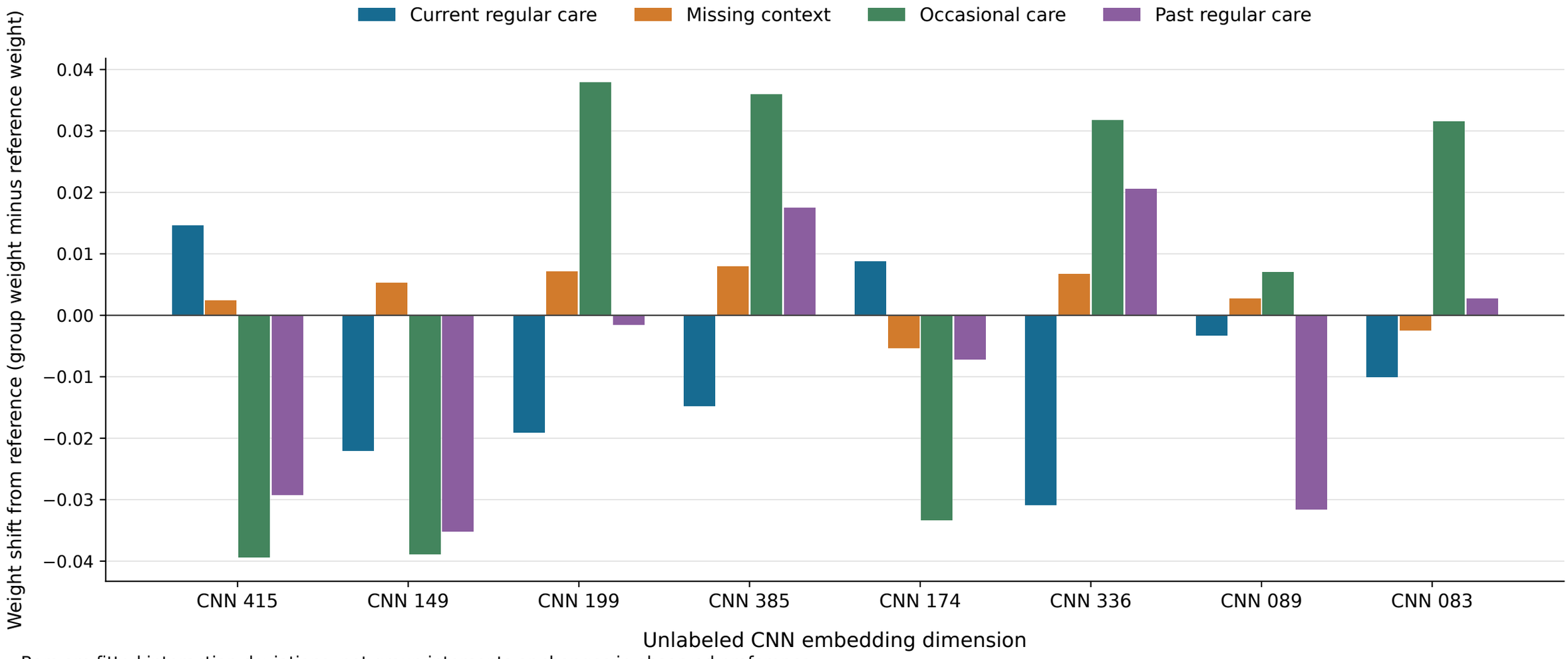
Numeric and categorical scales differ. 'Other' pools rare levels; CLIP-inferred breed is not true pedigree.

These are conditional model weights, not causal effects or standalone feature importance.

Pet-care context: shifts in CNN weights (M4-C)

Predictive Association, Not Causal Effect

Largest shifts across groups | Reference: Never | Reference shift = 0

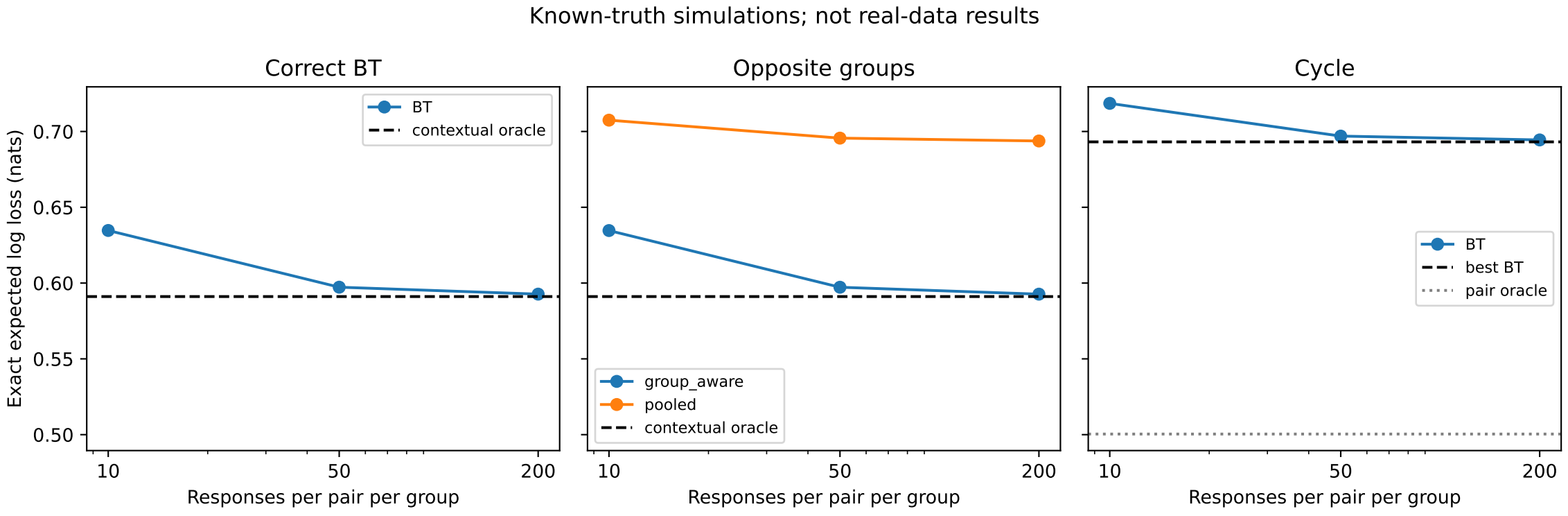


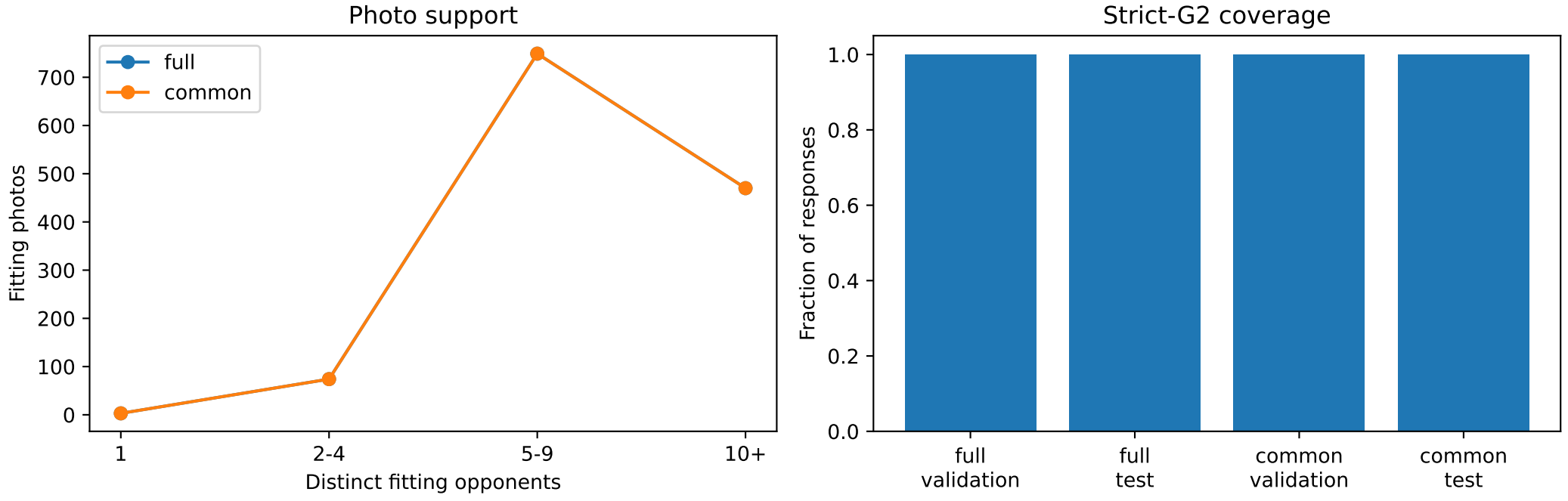
Bars are fitted interaction deviations, not group intercepts or changes in observed preference.

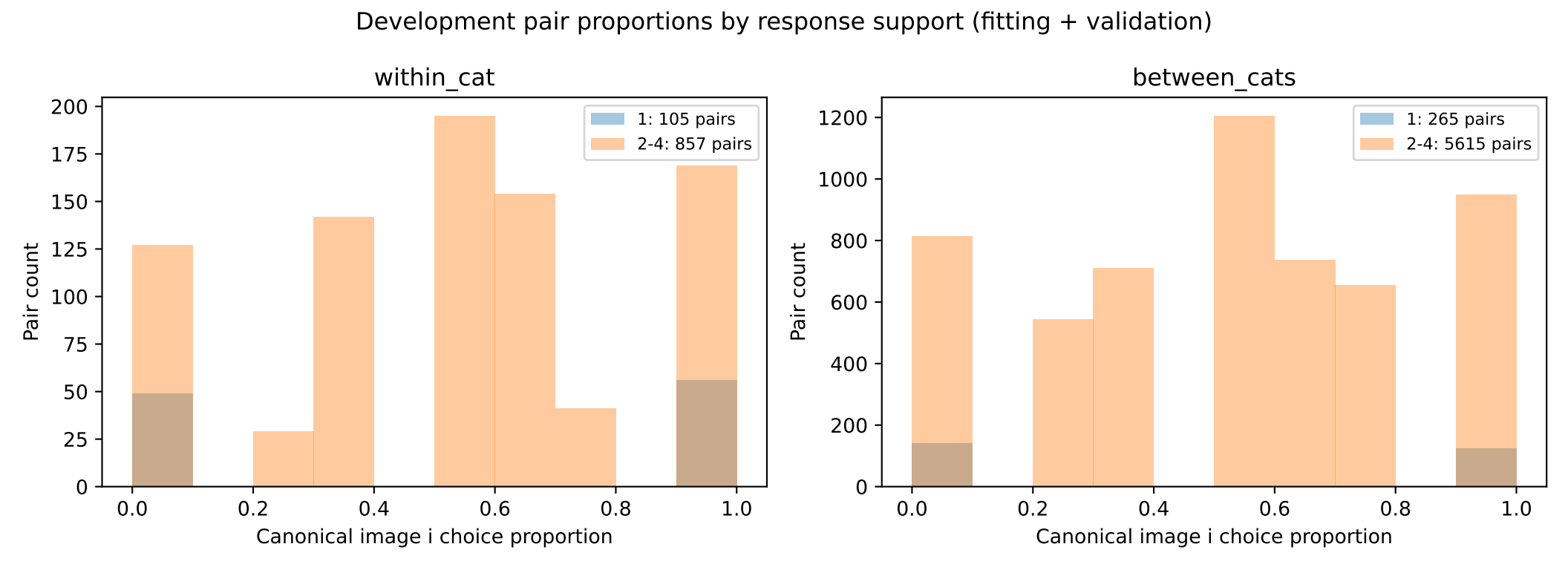
Shown: 8 dimensions ranked by maximum absolute deviation across the non-reference groups.

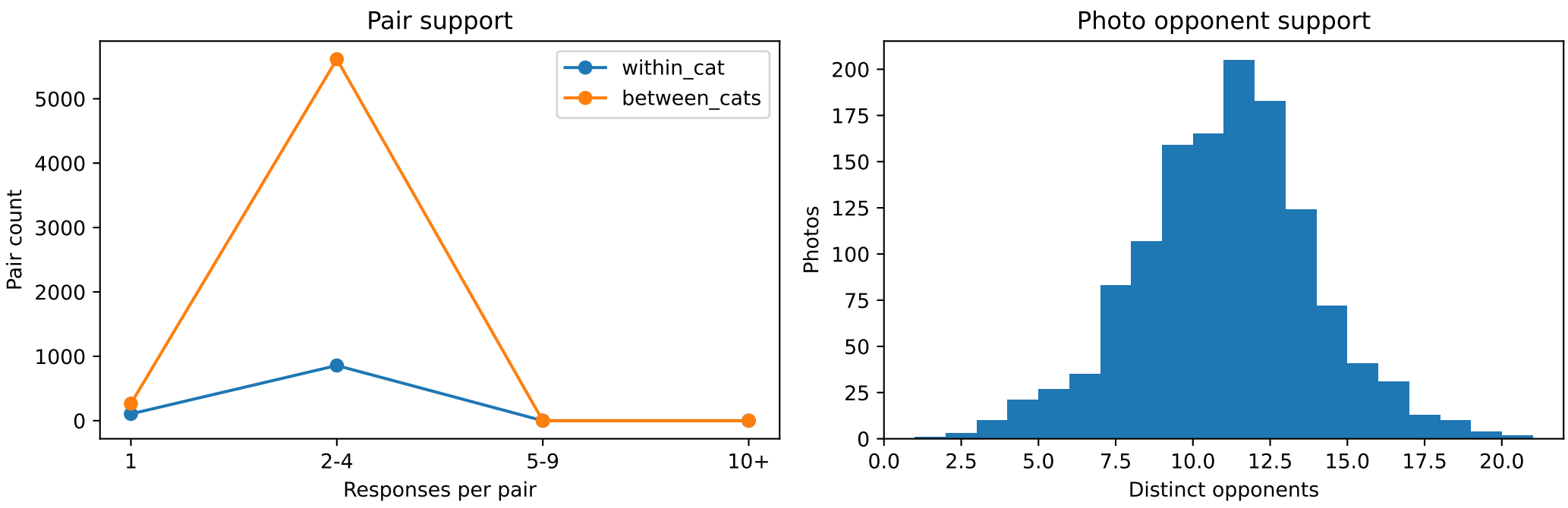
CNN coordinates have no individual semantic labels. Correlated coordinates and shrinkage affect their weights.

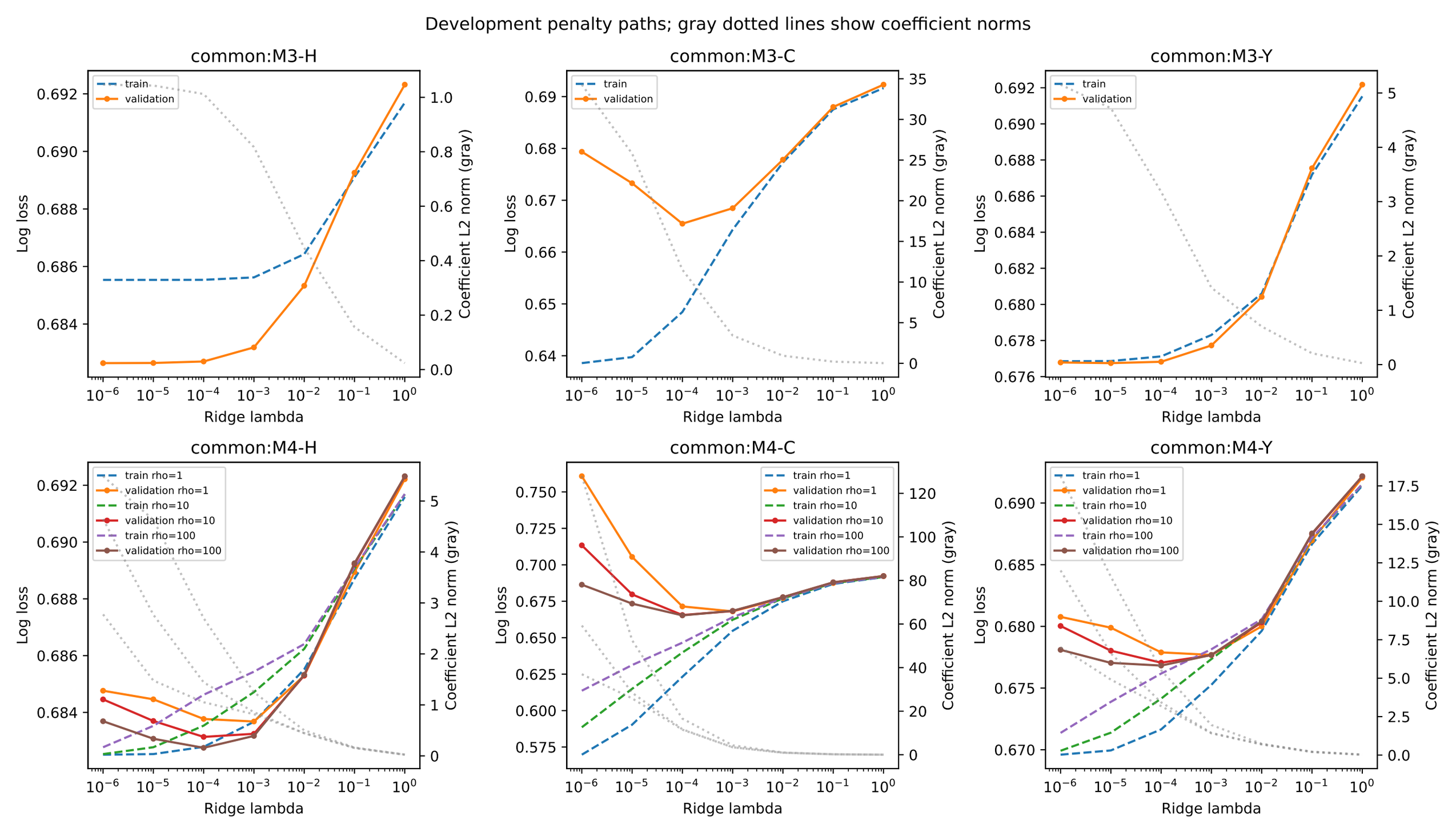
Descriptive model visualization after evaluation; not a new confirmatory test or causal interpretation.

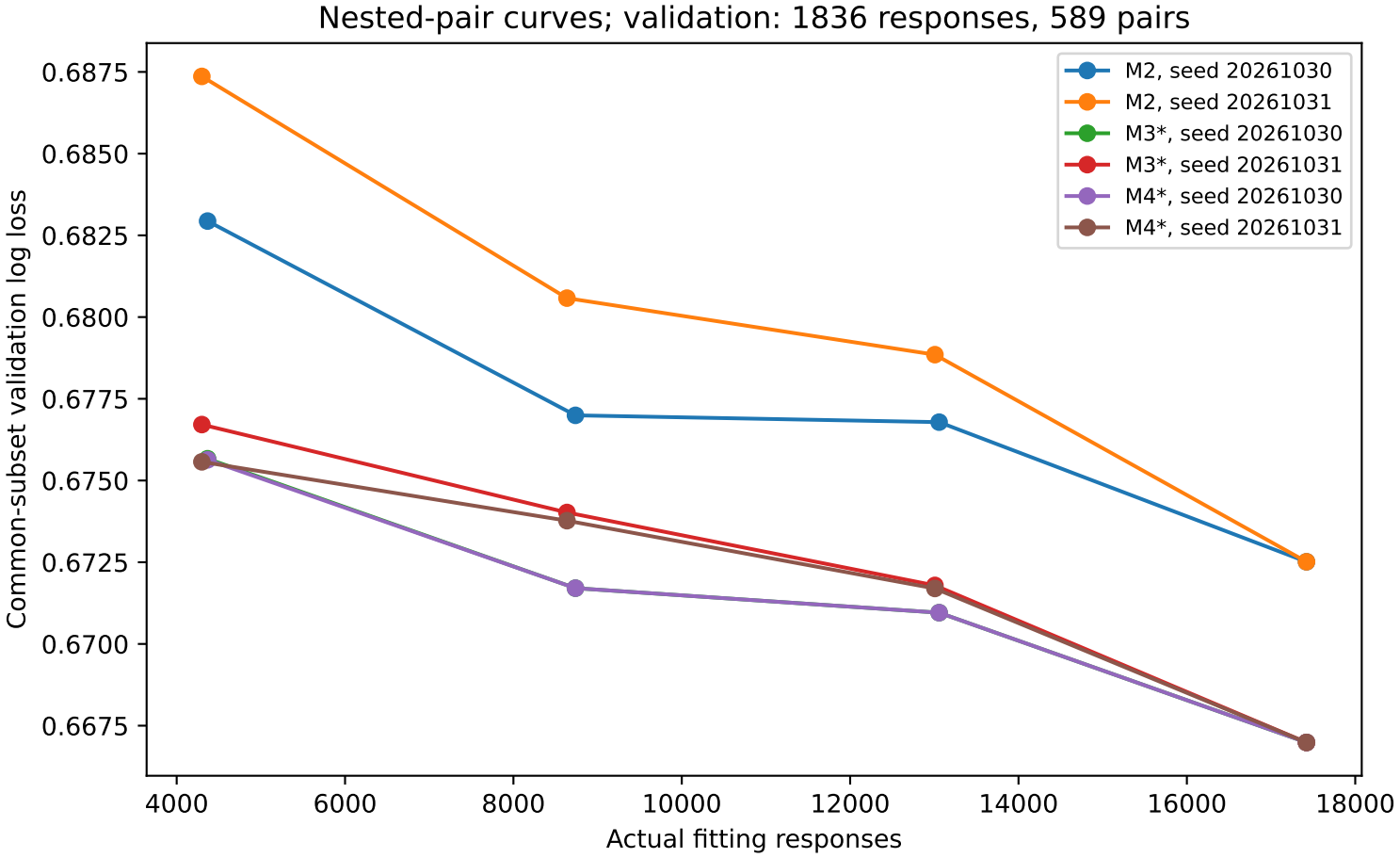

Known-truth simulations; not real-data results
Correct BT
Opposite groups
Cycle
Exact expected log loss (nats)
0.70
0.65
0.60
0.55
0.50
BT
contextual oracle
group_aware
pooled
best BT
pair oracle
10
50
200
Responses per pair per group

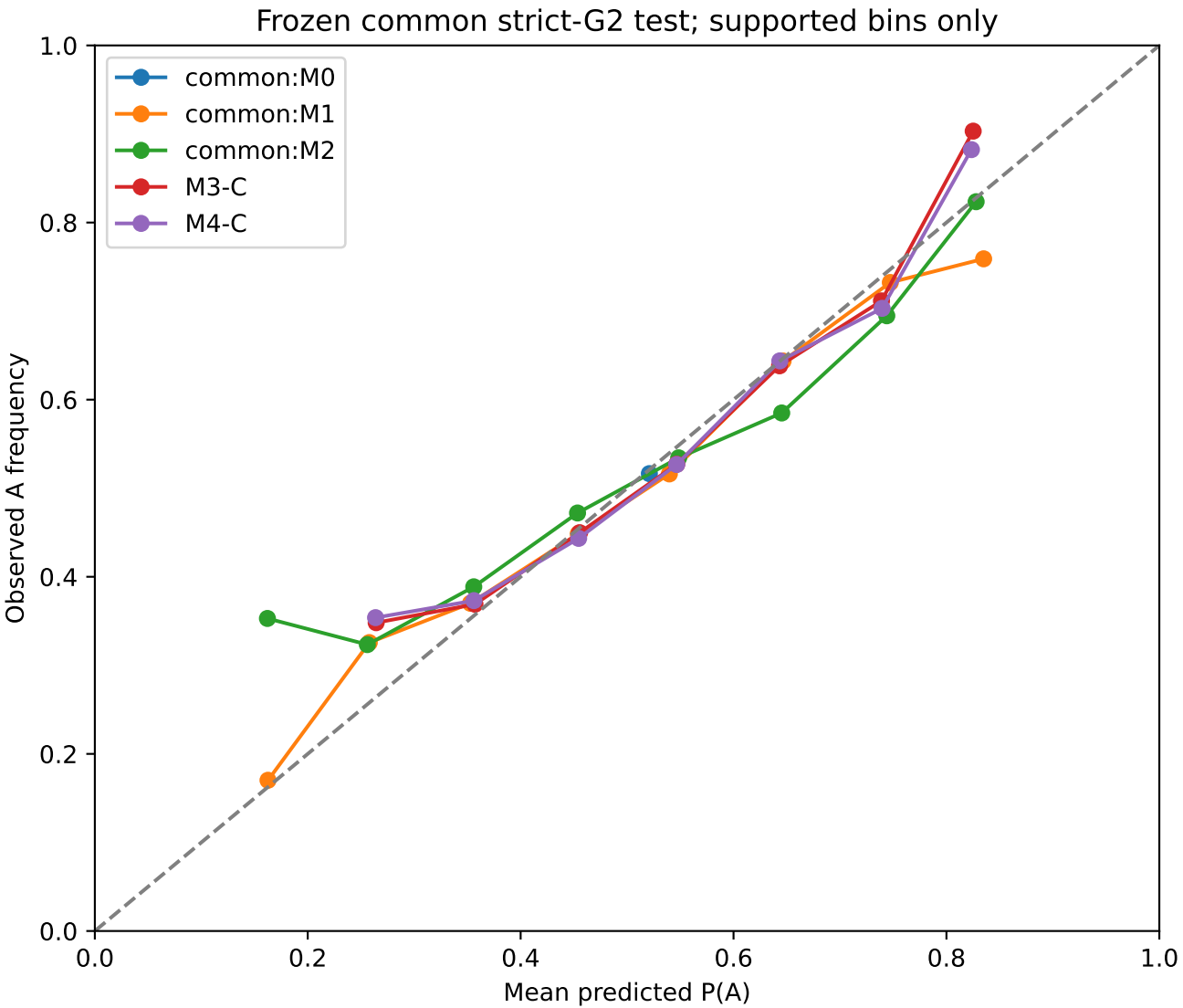

Fitting-graph support and structural held-out eligibility
Photo support
full
common
Fitting photos
0
100
200
300
400
500
600
700
1
2-4
5-9
10+
Distinct fitting opponents
Strict-G2 coverage
Fraction of responses
0.0
0.2
0.4
0.6
0.8
1.0
full validation
full test
common validation
common test

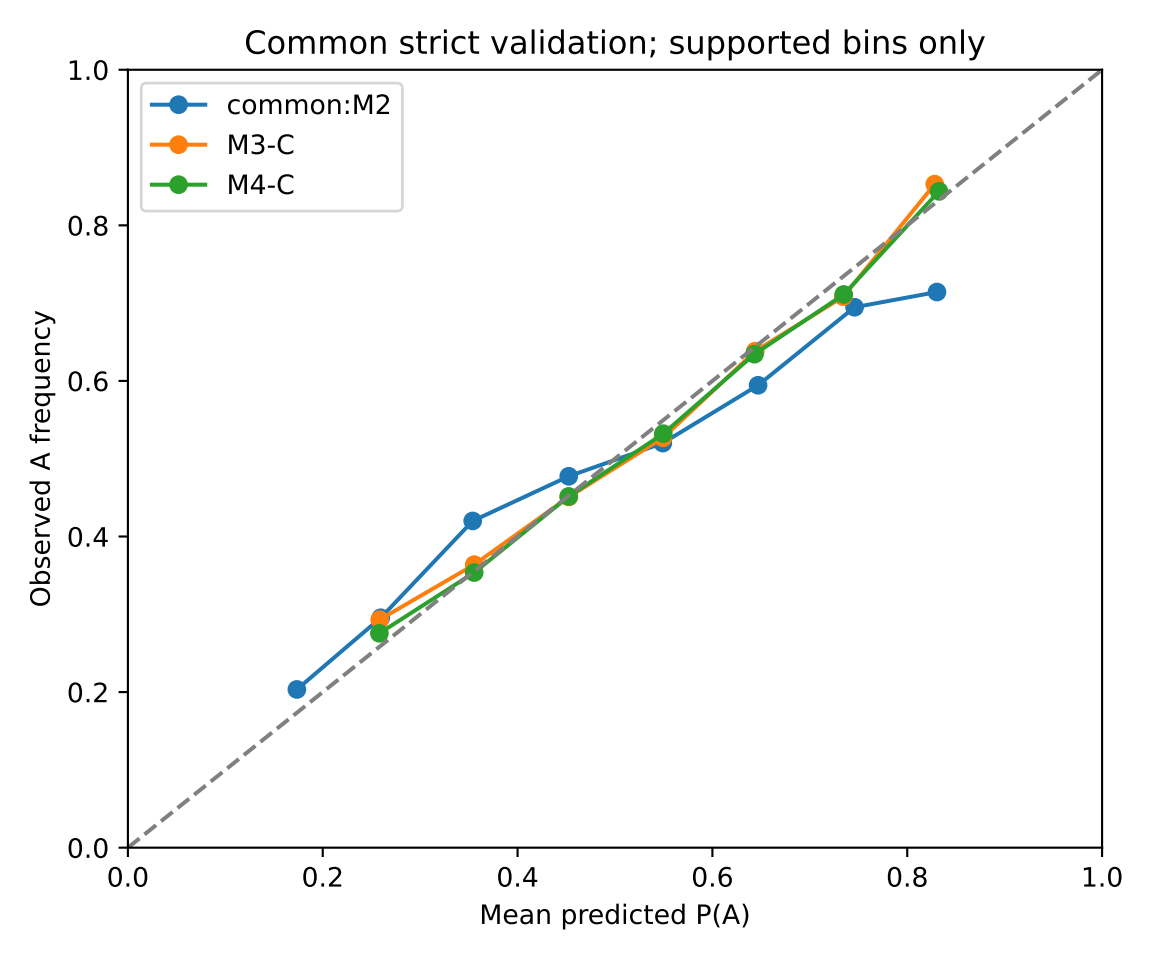

Development pair proportions by response support (fitting + validation)
within_cat
between_cats
1: 105 pairs
2-4: 857 pairs
1: 265 pairs
2-4: 5615 pairs
Pair count
Canonical image i choice proportion
0
25
50
75
100
125
150
175
200
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0


Development support (fitting + validation)
Pair support
within_cat
between_cats
Pair count
0
1000
2000
3000
4000
5000
1
2-4
5-9
10+
Responses per pair
Photo opponent support
Photos
0
25
50
75
100
125
150
175
200
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Distinct opponents


Development penalty paths; gray dotted lines show coefficient norms
common:M3-H
common:M3-C
common:M3-Y
common:M4-H
common:M4-C
common:M4-Y
train
validation
train rho=1
validation rho=1
train rho=10
validation rho=10
train rho=100
validation rho=100
Log loss
Coefficient L2 norm (gray)
Ridge lambda


Nested-pair curves; validation: 1836 responses, 589 pairs
Common-subset validation log loss
0.6875
0.6850
0.6825
0.6800
0.6775
0.6750
0.6725
0.6700
0.6675
4000
6000
8000
10000
12000
14000
16000
18000
Actual fitting responses
M2, seed 20261030
M2, seed 20261031
M3*, seed 20261030
M3*, seed 20261031
M4*, seed 20261030
M4*, seed 20261031

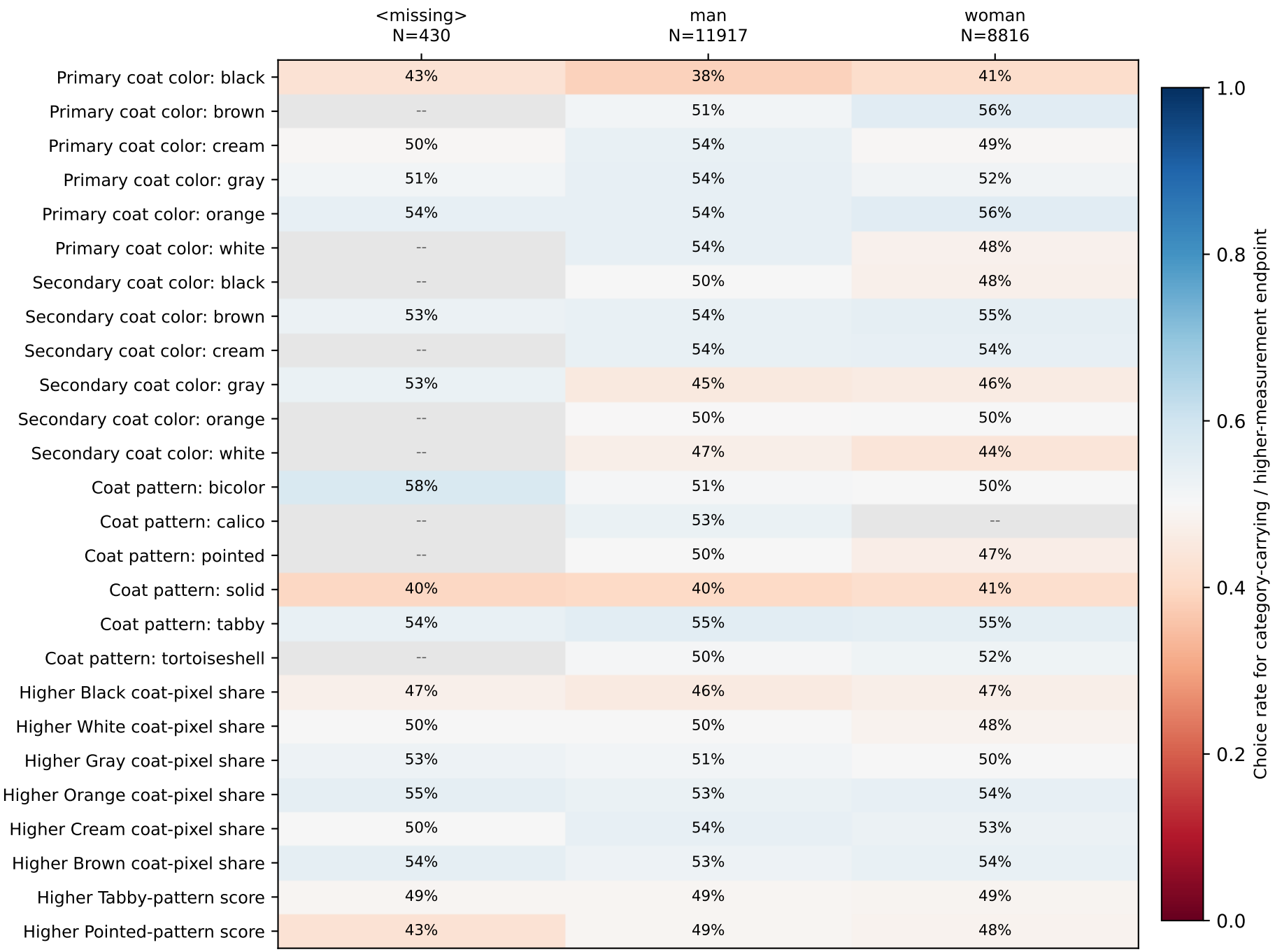

Frozen common strict-G2 test; supported bins only
common:M0
common:M1
common:M2
M3-C
M4-C
Observed A frequency
Mean predicted P(A)
0.0
0.2
0.4
0.6
0.8
1.0


Common strict validation; supported bins only
common:M2
M3-C
M4-C
Observed A frequency
Mean predicted P(A)
0.0
0.2
0.4
0.6
0.8
1.0

Gender and semantic image choices

Exploratory Observational Association, Not Causal Effect

Coat labels and measurements | Fitting + validation only



Only discriminating comparisons enter a rate. Gray cells: fewer than 100 responses or 30 pairs.

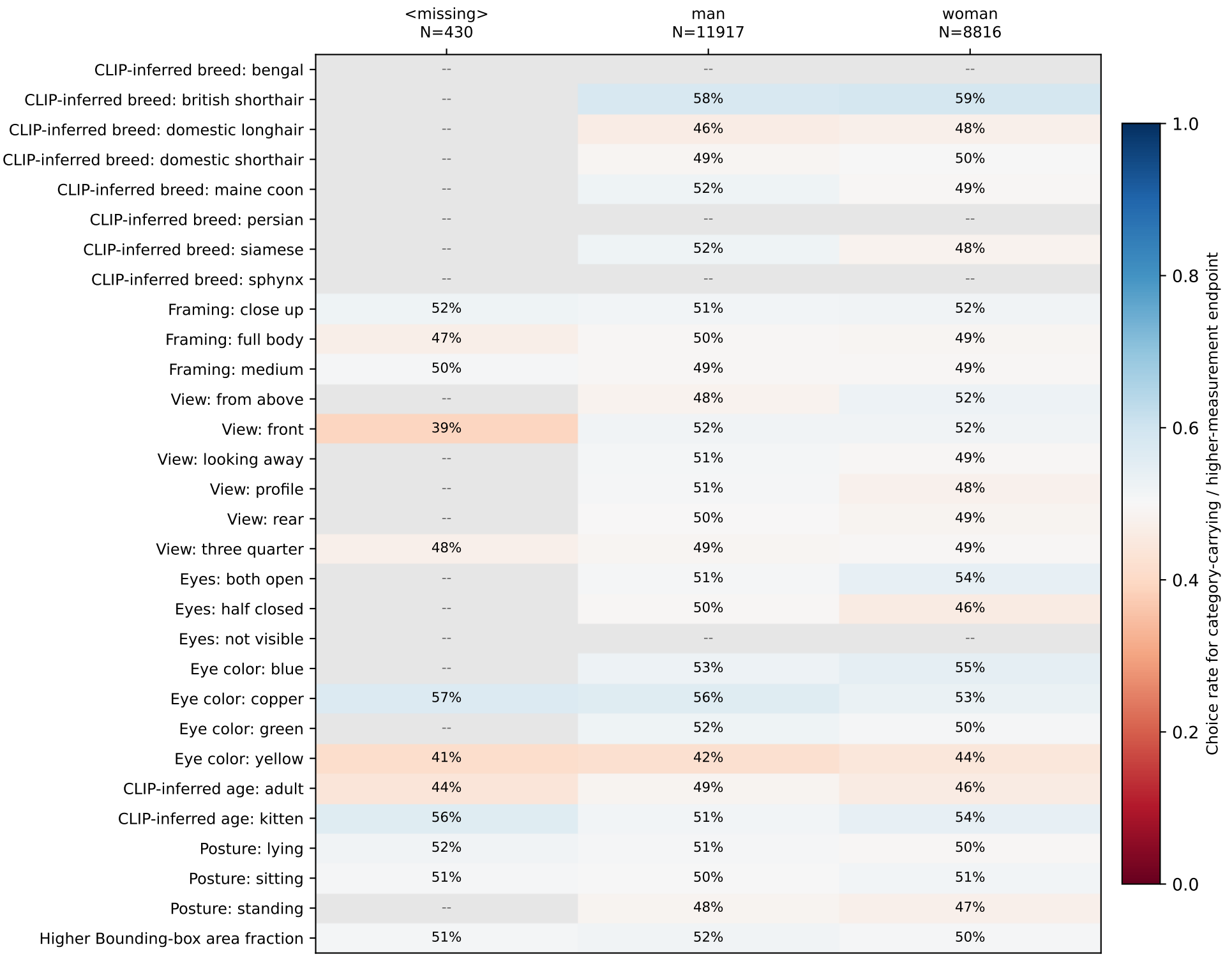
Groups may see different pairs. A rate above 50% is not an adjusted preference effect or a significance test.

Unknown semantic values and no-detection endpoints excluded. Dog fallbacks retained; CLIP-inferred breed is not true pedigree.

Gender and semantic image choices

Exploratory Observational Association, Not Causal Effect

Other automated attributes | Fitting + validation only



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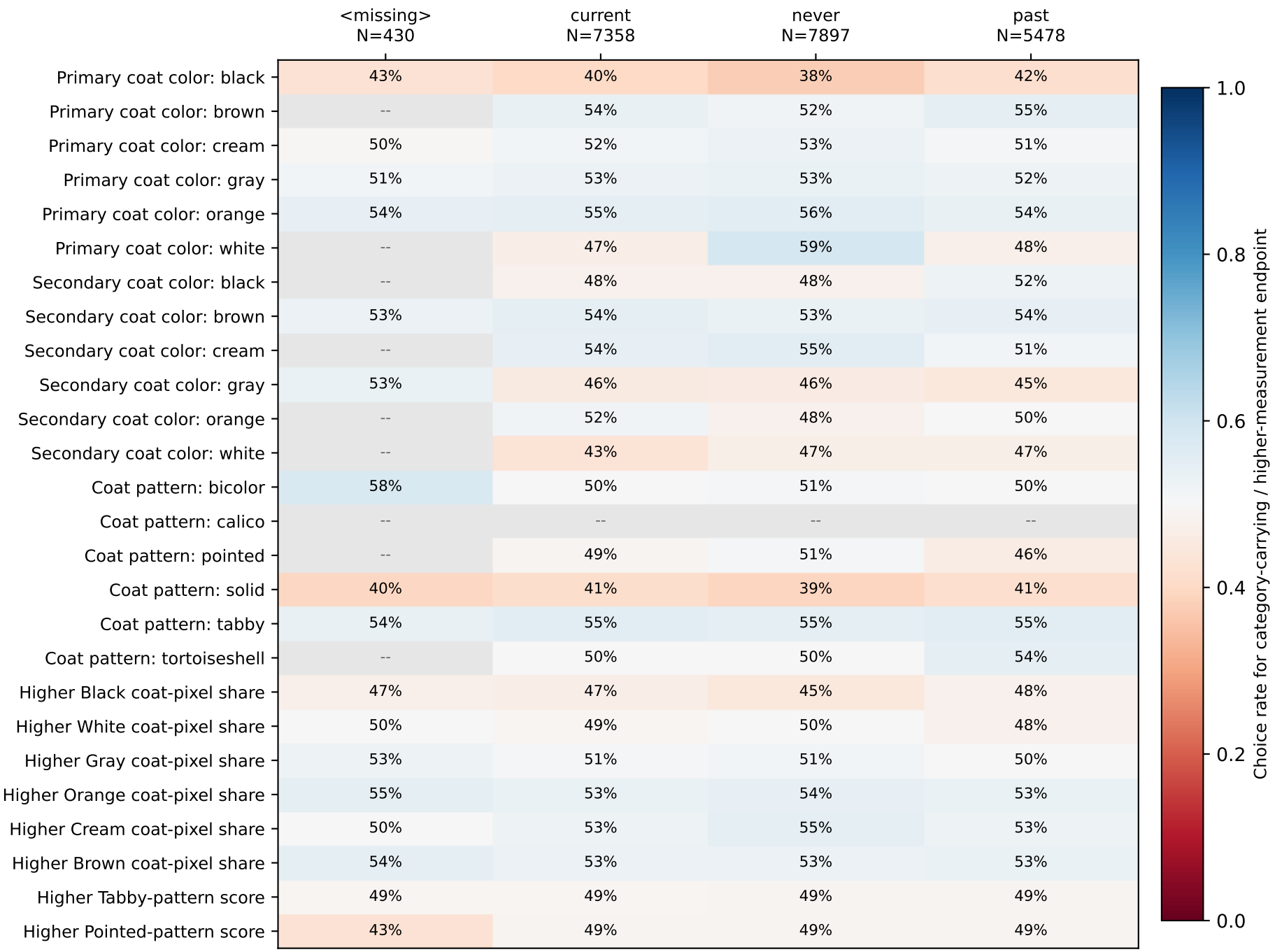
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Household pet history and semantic image choices

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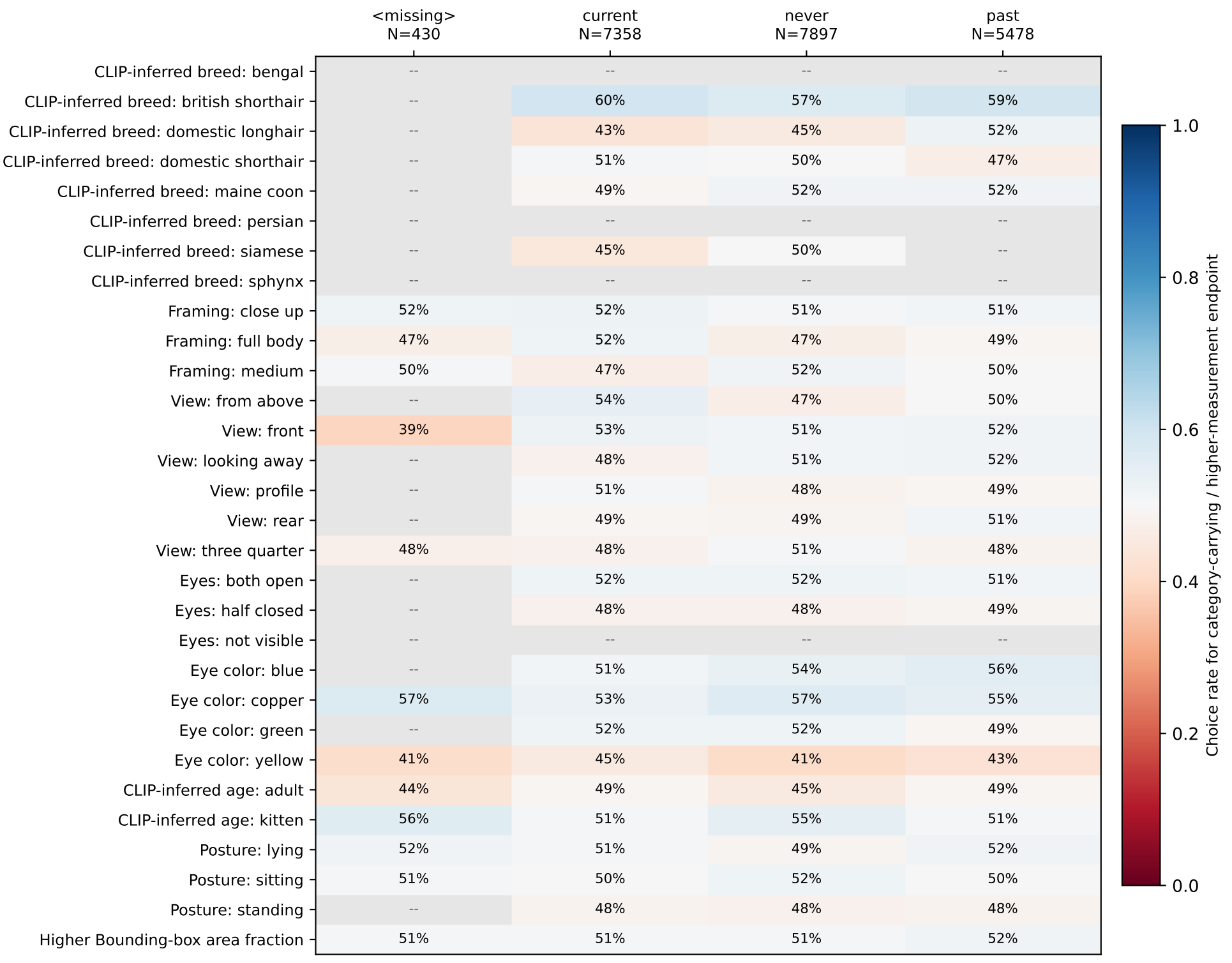
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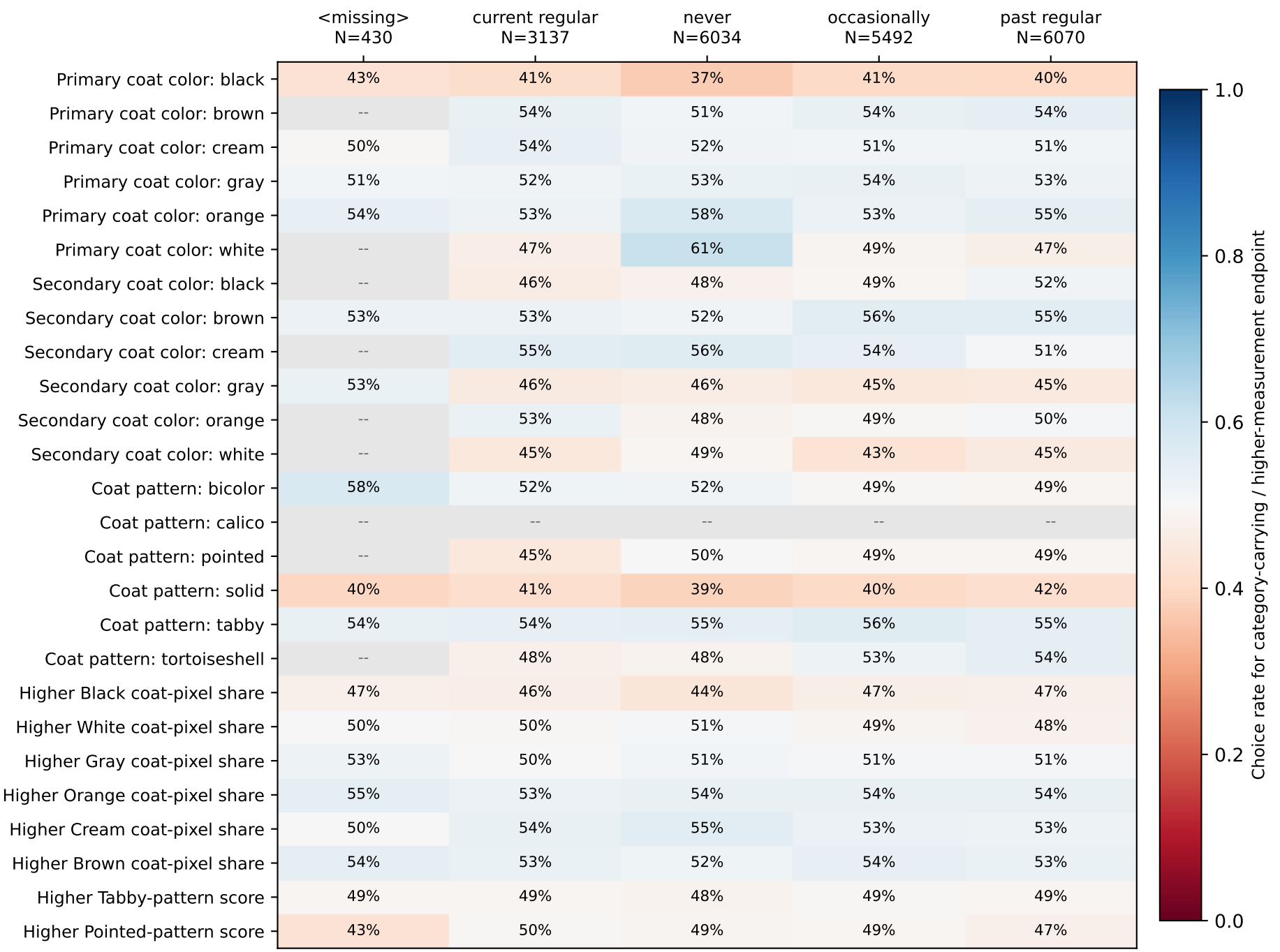
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Personal pet care and semantic image choices

Exploratory Observational Association, Not Causal Effect

Coat labels and measurements | Fitting + validation only



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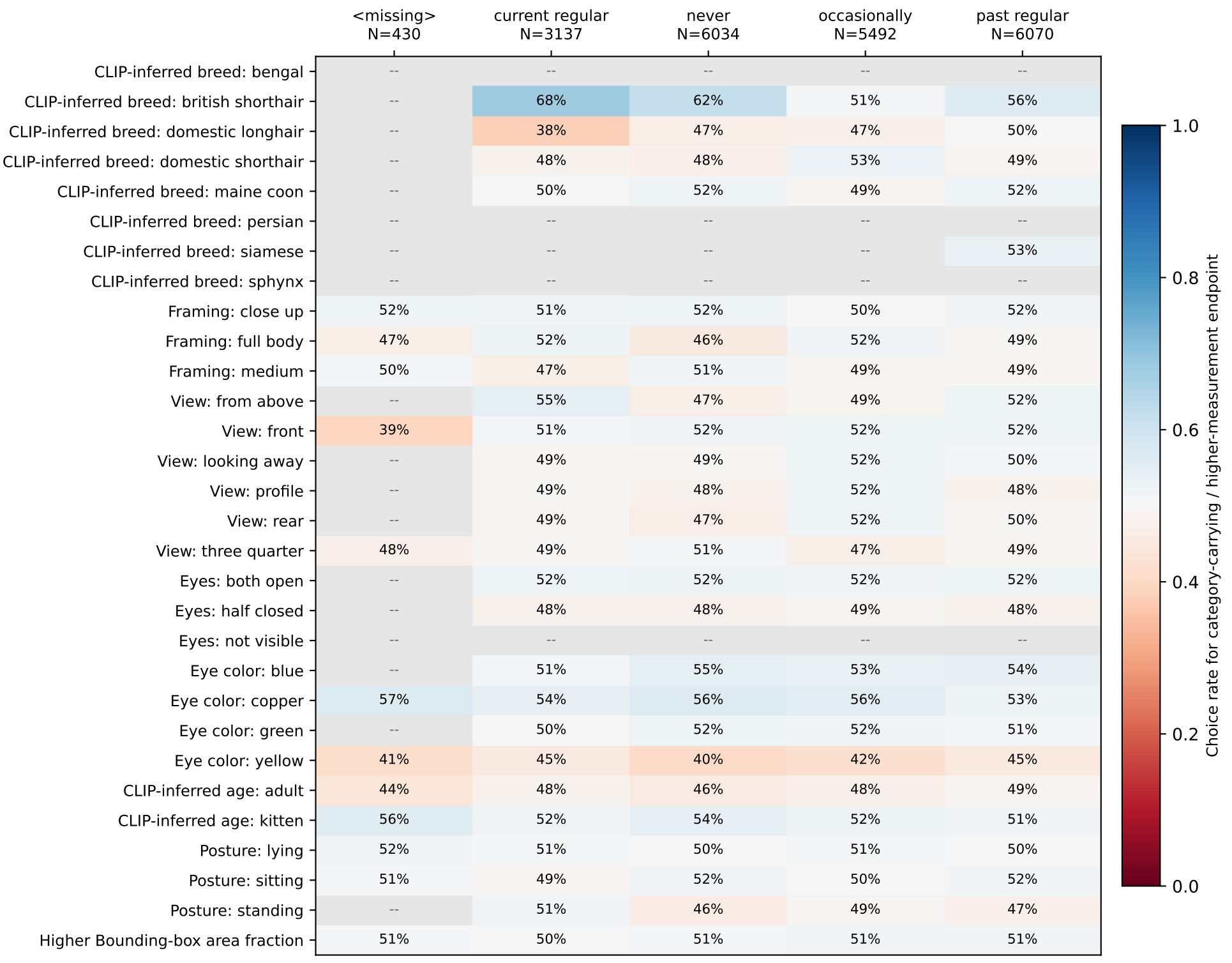
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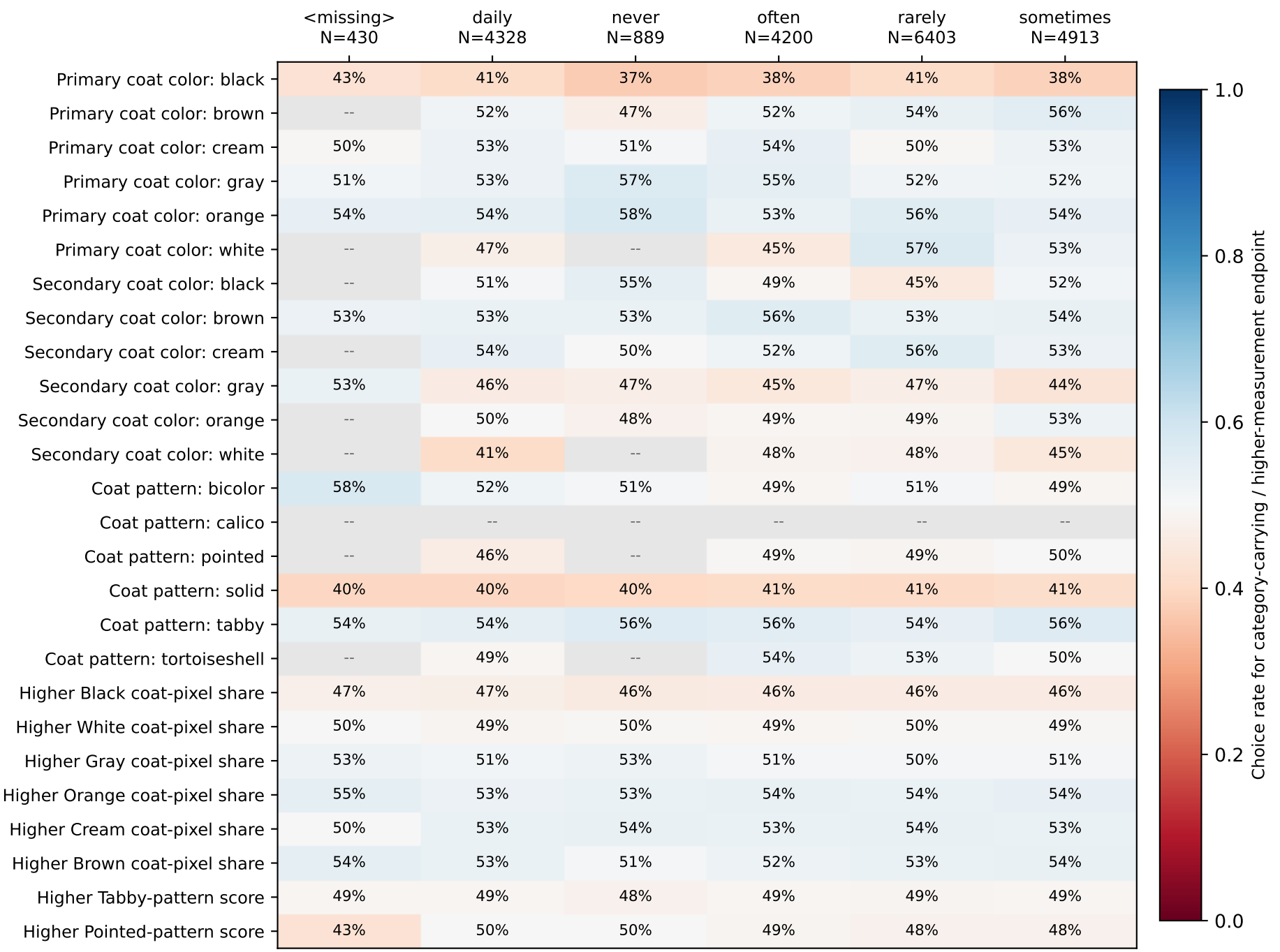
Groups may see different pairs. A rate above 50% is not an adjusted preference effect or a significance test.

Unknown semantic values and no-detection endpoints excluded. Dog fallbacks retained; CLIP-inferred breed is not true pedigree.

Pet contact frequency and semantic image choices

Exploratory Observational Association, Not Causal Effect

Coat labels and measurements | Fitting + validation only



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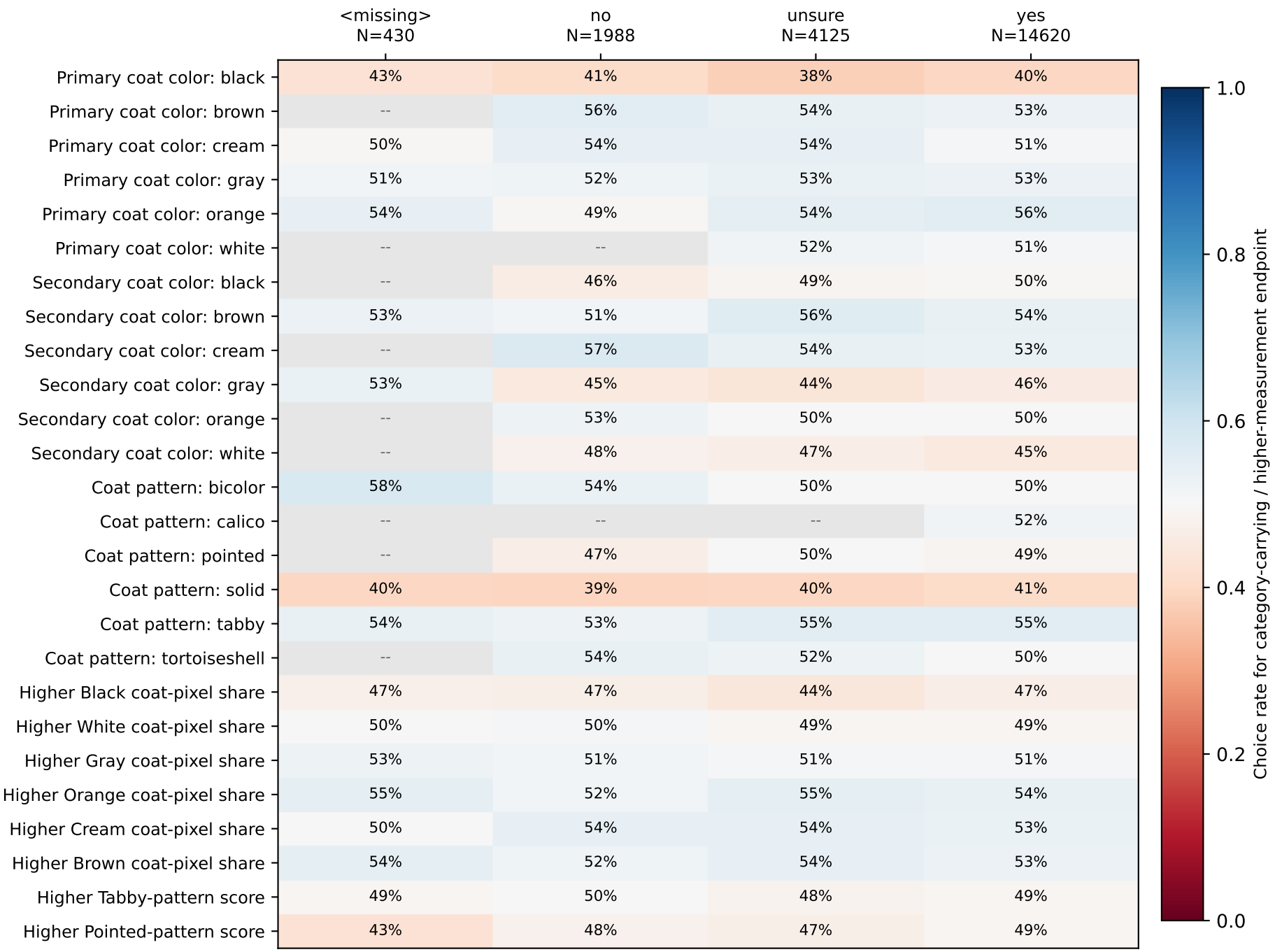
Groups may see different pairs. A rate above 50% is not an adjusted preference effect or a significance test.

Unknown semantic values and no-detection endpoints excluded. Dog fallbacks retained; CLIP-inferred breed is not true pedigree.

Future pet intention and semantic image choices

Exploratory Observational Association, Not Causal Effect

Coat labels and measurements | Fitting + validation only



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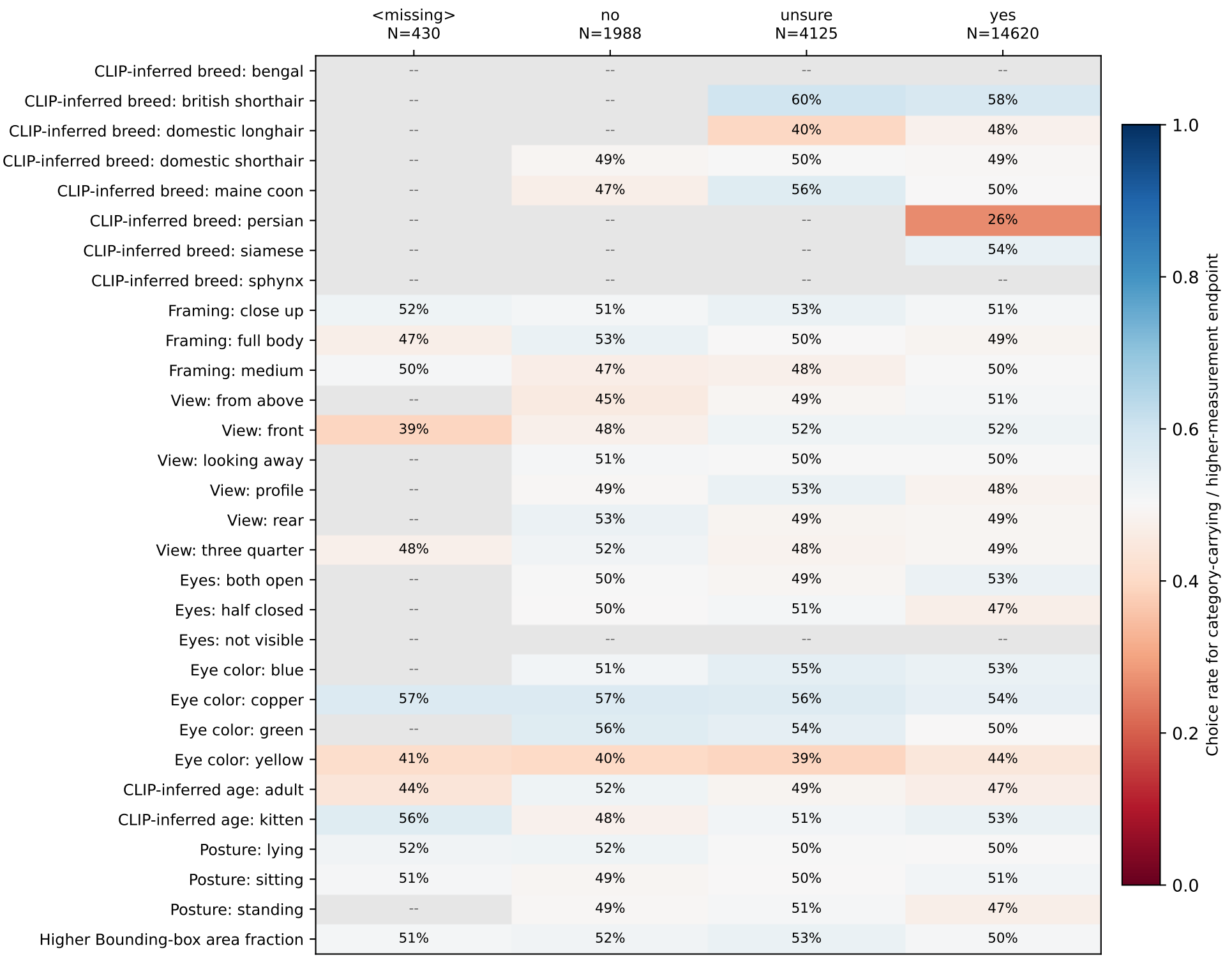
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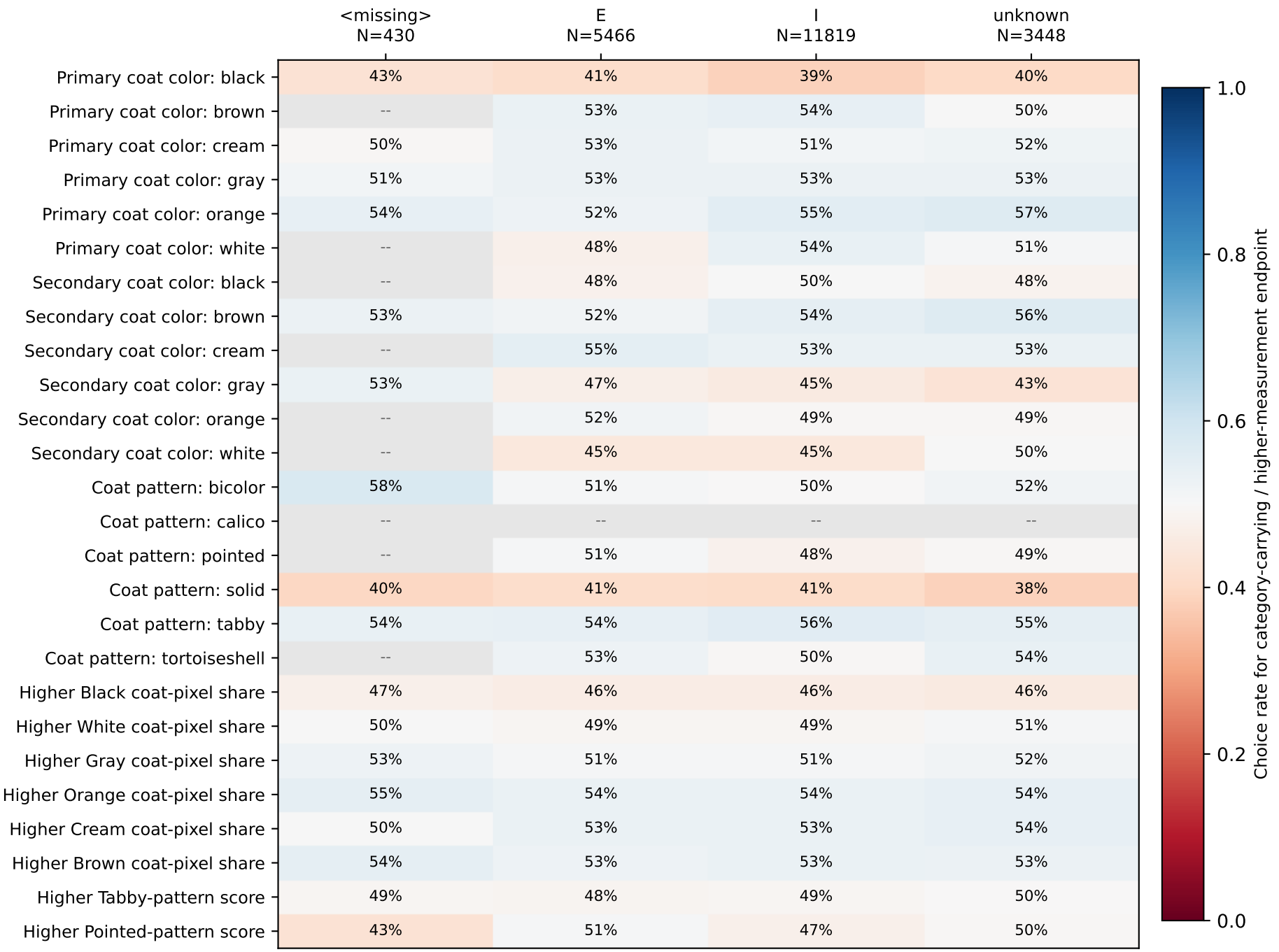
Groups may see different pairs. A rate above 50% is not an adjusted preference effect or a significance test.

Unknown semantic values and no-detection endpoints excluded. Dog fallbacks retained; CLIP-inferred breed is not true pedigree.

Self-reported MBTI I/E and semantic image choices

Exploratory Observational Association, Not Causal Effect

Coat labels and measurements | Fitting + validation only



Only discriminating comparisons enter a rate. Gray cells: fewer than 100 responses or 30 pairs.

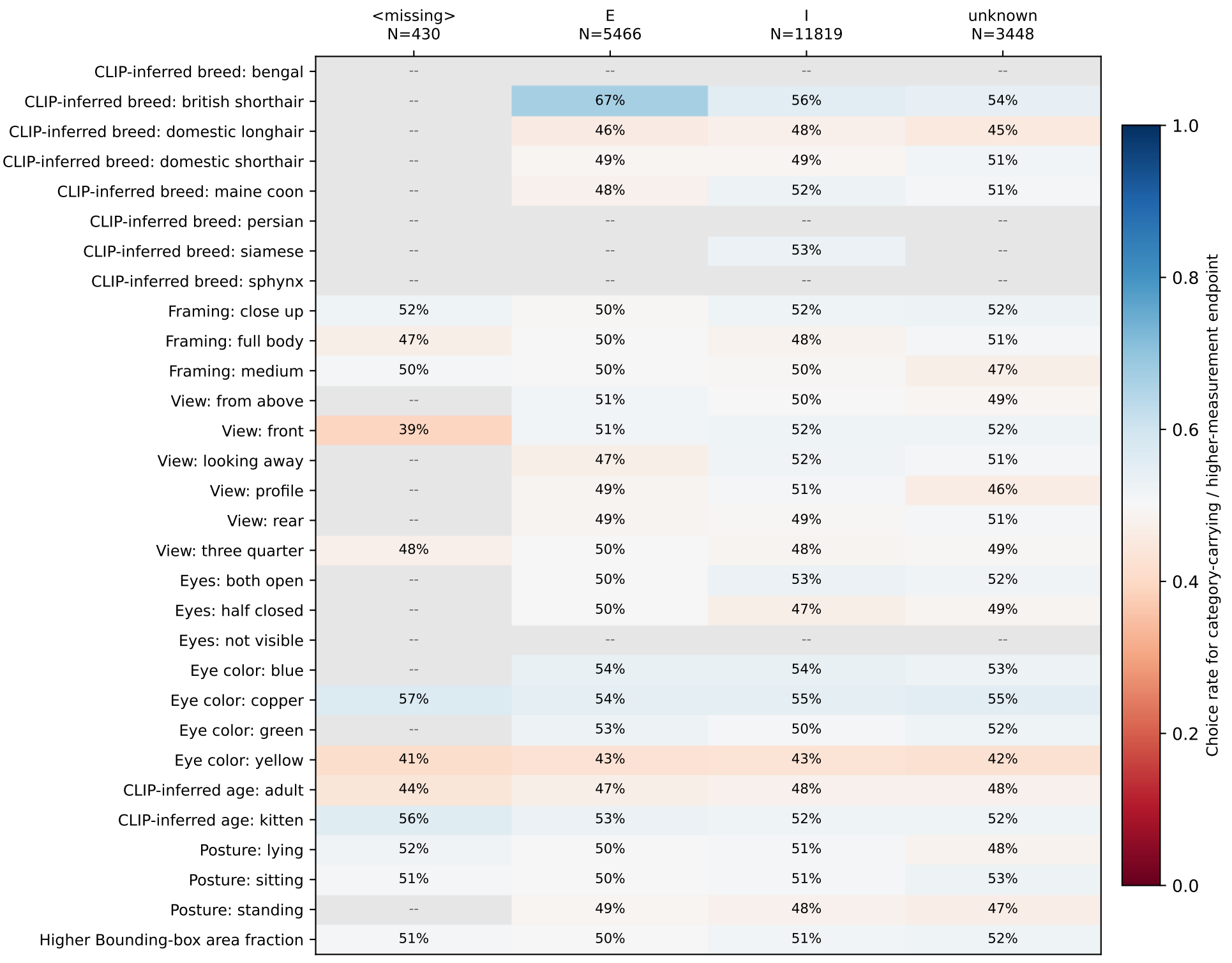
Groups may see different pairs. A rate above 50% is not an adjusted preference effect or a significance test.

Unknown semantic values and no-detection endpoints excluded. Dog fallbacks retained; CLIP-inferred breed is not true pedigree.

Self-reported MBTI I/E and semantic image choices

Exploratory Observational Association, Not Causal Effect

Other automated attributes | Fitting + validation only



Only discriminating comparisons enter a rate. Gray cells: fewer than 100 responses or 30 pairs.

Groups may see different pairs. A rate above 50% is not an adjusted preference effect or a significance test.

Unknown semantic values and no-detection endpoints excluded. Dog fallbacks retained; CLIP-inferred breed is not true pedigree.

Development demographic support | Exploratory Observational Association, Not Causal Effect

